

PES1UG24CS205

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CSE D SECTION

SCENARIO NO. 18 (SUBJECT RECRUITMENT ENGINE FOR CLINICAL TRIALS)

ID	Type	Description	Priority	Acceptance Criteria	Rationale
FR-001	Functional	The system shall evaluate candidate pre-screening responses against defined trial protocol inclusion/exclusion criteria.	High	Pass: Candidate failing ≥ 1 exclusion criteria is marked "Ineligible" with reasons logged. Fail: Ineligible candidates marked "Eligible" or failed criteria are omitted.	Prevents unqualified subjects from enrolling, protecting trial validity and patient safety.
FR-002	Functional	The system shall provide an electronic Informed Consent Form (eICF) workflow with digital signature capture and version tracking.	High	Pass: Timestamped signature captured; candidate status updates to "Consented"; PDF copy sent. Fail: Unsigned forms allow progression or	Compliance with GCP (Good Clinical Practice) and legal mandates for human subject trials.

				consent version is untracked.	
FR-003	Functional	The system shall enable Clinical Researchers to configure trial protocol criteria (demographics, biomarkers, medical history).	Medium	Pass: Researcher creates/updates criteria; updates reflect immediately in the evaluation engine. Fail: Criteria changes fail to persist or do not update real-time screening rules.	Protocols evolve across phases; researchers need dynamic rule configuration without code redeployment.
FR-004	Functional	The system shall log and track subject clinical trial visit compliance milestones (scheduled, completed, missed, rescheduled).	Medium	Pass: Researcher logs visit outcome; missed visit triggers an alert and compliance rate recalculation. Fail: Visit status updates fail to save or do not update overall subject compliance score.	Ensures protocol adherence and identifies non-compliant subjects risking study attrition.

FR-005	Functional	The system shall notify eligible candidates and assignees via automated SMS/email upon status change or upcoming visit.	Low	Pass: Notification dispatched within 60 seconds of eligibility classification or milestone trigger. Fail: Failed delivery without retry mechanism or incorrect subject notified.	Minimizes subject drop-off and maintains recruitment velocity.
NFR-001	Non-Functional	The system must comply with FDA 21 CFR Part 11 and HIPAA standards via immutable cryptographic audit trails and AES-256 encryption at rest.	High	Pass: All CRUD actions on subject data log an immutable entry (User ID, timestamp, prior/new value); zero plaintext PII. Fail: Missing audit entries for data modifications or plaintext storage of PHI.	Regulatory rejection and massive penalties occur if trial data integrity or patient privacy is breached.

NFR-002	Non-Functional	The eligibility screening engine must process and classify candidate surveys under peak load within sub-second latency, maintaining 99.9% uptime.	Medium	Pass: 95th percentile response time <800 ms under 500 concurrent survey submissions. Fail: p95>=800 ms or error rate >0.1% during load testing	Prevents candidate drop-off during high-traffic campaign recruitments and ensures trial continuity.
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Actors: Trial Candidate, Clinic Researcher, SMS notification, receptionist